

Highlights

Label-free covariance alignment for longitudinal EEG decoding

- A 24,181-parameter EEG decoder with label-free alignment that adapts at test time.
- Accuracy is level with eight published decoders trained in the same pipeline.
- In-network alignment reduces session drift by 46–63% in 15 of 15 subjects.
- Test-time adaptation can track the streaming class and erase the signal it protects.
- A lower bound on adaptation rate, read from the protocol, predicted a new cohort.